

5.2 Applications of Neural Networks and Deep Learning

نیورل نیٹ ورکس اور ڈیپ لرننگ کی مشہور ایپلی کیشنز یہ ہیں: تصویر کی پہچان، آواز کی پہچان، قدرتی زبان کی پروسیس...

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 08 of 17

5.2 Applications of Neural Networks and Deep Learning

LEARN IT

Neural networks and deep learning are widely used in modern life. Many intelligent systems use them in the background to **recognise information, make predictions and support users**.

#	Application	What it does	Example
1	Image recognition	Identifies objects, faces or patterns in images. Used in security cameras, photo search and biometric locks.	Unlocking a smartphone using face recognition; a camera at a Karachi plaza matching a face with a staff list.
2	Speech recognition	Converts spoken words into digital text or commands. Deep learning helps with accents, pronunciation and voice patterns.	A student speaks to Google Assistant or Alexa to set a morning alarm; Urdu voice search on a phone.
3	Natural Language Processing (NLP)	Helps computers understand, process and respond to human language grammatically and in context. Core of chatbots, translators and AI writers.	Using Google Translate to turn a difficult English paragraph into Urdu; a bank chatbot answering in Urdu.
4	Healthcare systems	Analyse complex medical images (X-rays, CT scans, MRIs) to detect diseases early and accurately.	A hospital system highlighting abnormal patterns in a lung X-ray that might indicate pneumonia or tumours — useful at a DHQ hospital that has few radiologists.
5	Recommendation systems	Analyse past behaviour and preferences to suggest content, products or services.	A student watches computer-science videos on YouTube; the platform recommends more programming tutorials.
6	Autonomous vehicles	Self-driving cars use neural networks to process real-time camera and radar data: roads, signals, pedestrians, obstacles.	A self-driving car detects a pedestrian crossing and automatically applies the brakes.
7	Fraud detection	Banks compare current activity with previous spending patterns to catch unusual transactions.	An HBL or Meezan app flags a sudden Rs 80,000 payment from a new city at 3 a.m.
8	Assistive technologies	AI supports people with disabilities.	Speech-to-text tools for users who have difficulty typing; screen readers that speak Urdu text aloud.

WORKED EXAMPLE

Worked example: match the app to the application name — a day in Ahmed's life in Hyderabad.

- 7:00 am — phone unlocks by face. → **Image recognition**.
- 7:05 am — "Ok Google, set alarm for Maghrib." → **Speech recognition**.
- 8:00 am — Google Translate helps with an English article. → **NLP**.
- 4:00 pm — YouTube suggests a Python tutorial after he watched Chapter 5 notes. → **Recommendation system**.
- 9:00 pm — JazzCash texts "We blocked a payment that does not match your pattern." → **Fraud detection**.

REMEMBER IT

Eight applications: I-S-N-H-R-A-F-A. Image, Speech, NLP, Healthcare, Recommendations, Autonomous vehicles, Fraud, Assistive tech. "I See New Hospitals Recommend Autonomous Fraud-Assist."

COMMON MISTAKE

Mistake: Mixing up speech recognition and NLP.

Fix: Speech recognition turns **sound into text** (or a command). NLP understands the **meaning of language** (grammar, context, translation, chat replies). You can have NLP on typed text with no microphone at all.

اردو میں سمجھیں

نیورل نیٹ ورکس اور ڈیپ لرننگ کی مشہور ایپلی کیشنز یہ ہیں: تصویر کی پہچان، آواز کی پہچان، قدرتی زبان کی پروسیسنگ (NLP)، صحت کے نظام، تجویز کرنے والے نظام، خود چلنے والی گاڑیاں، فراڈ کی نشاندہی اور معاون ٹیکنالوجیز۔ روزمرہ میں چہرے سے فون کھولنا، یوٹیوب کی تجاویز اور بینک کا مشکوک لین دین پکڑنا سب اسی خاندان سے ہیں۔

CHECK YOURSELF

- Which application converts spoken words into text or commands?
- Give the source example of a recommendation system.
- How do self-driving cars use neural networks?

Answers: 1. Speech recognition. 2. YouTube recommending more programming tutorials after a student watches computer-science videos. 3. They process real-time camera and radar data to understand roads, traffic signals, pedestrians and obstacles (for example braking for a pedestrian).

Next lecture: 5.3.1 Secure Collaboration (Authentication, MFA, Access Control)